

Lexicographic Minimax Regret over Declared Monotone Probes for Robust Multicriteria Selection

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Abstract

The efficient set is a safety filter, not a complete decision rule. To provide a principled recommendation when the decision maker cannot state precise weights, we propose the Leximax Universal-Regret (LexUR) core. LexUR evaluates alternatives over a declared family of monotone questions (probes). It ranks candidates by the lexicographically sorted vector of their dimensionless regrets, returning a robust stability class accompanied by an interpretable certificate listing the binding questions.

We prove existence and completeness, Pareto compatibility, tolerance-rule stability under bounded perturbations, and a probe perturbation bound. We also provide an adaptive correlation-clustering construction that reduces sensitivity to redundant criteria.

In a comprehensive audit protocol spanning 11 methods, 10 preference families, and 8 front geometries, LexUR is competitive with the strongest robust scalarisation baselines on tail held-out loss. It remains non-inferior to ASF and sampled minimax regret under paired tests, while a single-point hypervolume score obtains the best average rank. LexUR's distinct contributions are its auditable regret certificate and reduced redundancy bias relative to averaging methods. Because point-estimate normalisation can be sensitive, robust deployment should report an interval stability class. A unique recommendation is returned only when margin conditions hold. LexUR is robust with respect to the declared probe family, not all possible preferences. We also outline an exploratory stochastic extension illustrated on a synthetic smart-grid economic dispatch model.

Keywords: Decision support, Multicriteria decision analysis, Robust decision making, Minimax regret, Achievement scalarising functions, Pareto efficiency

1. Introduction

Since the work of Pareto, efficiency has been the cornerstone of multicriteria analysis: an alternative is Pareto efficient if no criterion can be improved without worsening another. This is a safety property that removes dominated alternatives, but it is decisionally incomplete. For m conflicting criteria the efficient set is typically large, leaving the decision maker (DM) with no principled selection rule. The fundamental issue remains that the Pareto front is a safety filter, not a complete decision rule.

LexUR turns the output from a set into a certified recommendation. The DM declares a monotone probe family and normalization bounds. Each alternative receives normalized regrets over that family. These regrets are lexicographically sorted, and selection follows a lexicographic minimax rule. This yields a certified stability class accompanied by an explanation of which questions remain most severely binding. Figure 1 contrasts the safety filter with the regret certificate.

Our contributions are unified across four dimensions. Theoretically, we define the LexUR order and prove existence, completeness, Pareto compatibility, and stability under bounded perturbations. Algorithmically, we provide an exact finite-set procedure using sequential sorts. Empirically, we validate LexUR on a broadened benchmark suite, demonstrating competitive tail performance against strong

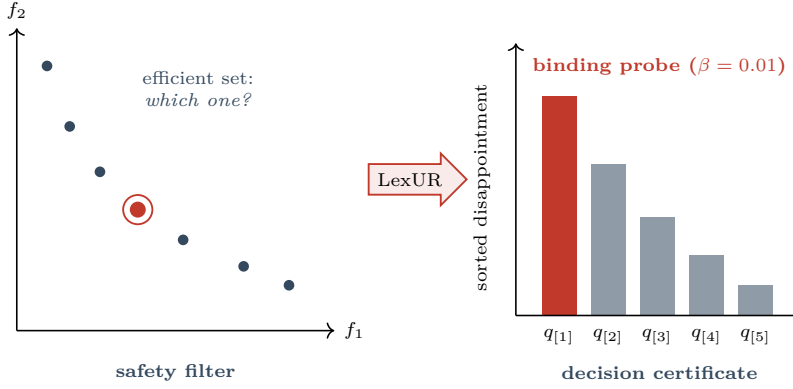


Figure 1: From a front to a certificate. The efficient set (left) removes dominated options but leaves the choice open. LexUR returns a recommendation or stability class together with an illustrative sorted regret certificate (right). A probe is called binding only when its disappointment lies within the declared presentation band $\beta = 0.01$ of the maximum; the remaining bars are secondary lexicographic constraints.

robust baselines. Interpretably, we introduce an adaptive correlation-clustering probe family that handles redundancy and produces an auditable regret certificate for the final choice.

Robustness in this framework is strictly relative to the declared probe family, not all possible preferences. Furthermore, point normalization may be sensitive when ideal and nadir bounds are uncertain. LexUR is not parameter-free, but its assumptions are declared, auditable, and separated from the final recommendation.

Section 2 positions the method. Sections 3 to 5 develop the core theory, computation, and probe design. Sections 6 and 7 test extensions and empirical performance, and Section 8 closes with implications.

2. Background and related work

Table 1 compactly maps LexUR against the standard families of multicriteria decision analysis. Broadly, existing selection devices fall into four categories: utility aggregation (e.g., weighted sum, OWA [13, 11]), distance to reference rules (e.g., TOPSIS [5], CP [14], ASF [12]), regret-based rules [9, 6], and stochastic acceptability or robustness frameworks. The classical methods are robust but typically require elicited weights, thresholds, metrics, or explicit reference points, as detailed in the comparative table.

LexUR aligns most closely with robust decision analysis, particularly SMAA [7, 10], robust ordinal regression (ROR) [3, 2], and preference robust optimisation (PRO) [1, 4]. However, LexUR’s operational mechanics and outputs are distinct. Rather than returning an acceptability index over a weight distribution (SMAA), reasoning over elicited value functions (ROR), or optimising a worst-case expected utility (PRO), LexUR optimises the lexicographically sorted vector of normalised regrets. Crucially, it operates over an explicit, declared family of monotone questions (probes) and outputs an auditable certificate that names the binding coalitions without requiring elicited preferences or sampling assumptions.

LexUR does not subsume or replace Pareto analysis. It is a prescriptive rule that turns a feasible candidate set into a robust, explainable recommendation under declared monotone probes. It recovers specific methods like minimax regret or ASF only under restricted, singleton probe families. Ultimately, its distinguishing contribution is formulating decision-making as a lexicographic minimax over normalised disappointments, supplemented by an adaptive redundancy-aware probe construction and an interpretable regret certificate.

Table 1: Comparison of LexUR with representative MCDA and robust selection methods.

Method	Required Input	Pareto Comp.	Output / Explanation	choice	Key Limitation
Weighted sum	Weight vector	Yes ($w > 0$)	Single (Low)	choice	Misses non-supported points in non-convex fronts
TOPSIS	Weights, distance metric	Not strictly	Single (Low)	choice	Rank reversal, distance to non-feasible ideal
CP	Weights, L_p exponent	Yes ($p < \infty$)	Single (Low)	choice	Non-linear scaling effects
ASF / Chebyshev	Reference point, scaling	Yes	Single (Low)	choice	Requires an arbitrary reference point
MMR (sampled)	Linear weight distribution	Yes	Worst-case regret		Limited to linear aggregations
SMAA	Preference distribution	Usually	Acceptability indices		Exploratory, single-choice depends on post-processing
VIKOR	Weights, majority threshold	Not strictly	Single (Low)	choice	Complex parameter interaction
Single-point HV	Reference point	Yes	Single (Low)	choice	Box-volume score, sensitive to reference point
LexUR (ours)	Monotone probe family	Yes*	Leximax certificate	certifi-	Sensitive to normalisation bounds

* Strict Pareto compatibility holds if probes separate criteria (Theorem 2).

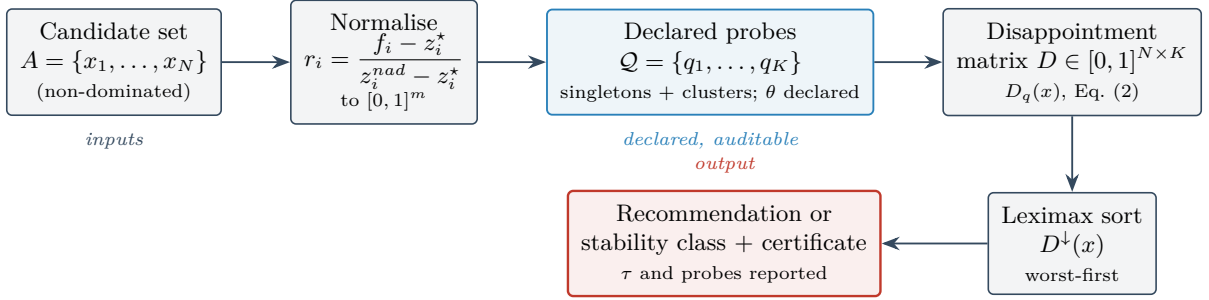


Figure 2: The LexUR pipeline. Declared inputs include the candidate set, normalisation bounds, probe family $\mathcal{Q}$, clustering rule θ , and numerical tolerance τ . Exact leximax comparison uses $\tau = 0$; numerical reporting returns a stability class unless a verified margin supports a unique recommendation. Every output retains the labelled regret certificate.

3. The Leximax Universal-Regret core

Figure 2 illustrates the LexUR rule: from a candidate set, through normalisation and the declared probe family, to the leximax selection and its certificate.

3.1. Problem setting and normalisation

Let $\mathcal{X} \subseteq \mathbb{R}^n$ be the feasible set and $F : \mathcal{X} \rightarrow \mathbb{R}^m$ the vector objective to be minimised. We consider a candidate set $A = \{x_1, \dots, x_N\} \subseteq \mathcal{X}$. Criteria are normalised to $[0, 1]$ by the positive affine map

$$r_i(x) = \frac{f_i(x) - z_i^*}{z_i^{\text{nad}} - z_i^*}, \quad z_i^* = \min_{x \in A} f_i(x), \quad z_i^{\text{nad}} = \max_{x \in A} f_i(x), \quad (1)$$

which attains both endpoints on A . Degenerate criteria (where $z_i^{\text{nad}} - z_i^* \approx 0$) are excluded from non-singleton probe construction unless external bounds are provided. Because active-set normalisation is set-dependent, LexUR should be used with externally fixed bounds or accompanied by a stability analysis.



B is recommended: its *worst* disappointment is smallest. On a tie at the worst entry, the next sorted coordinate decides, and so on.

Figure 3: The lexicimax order. Each alternative’s probe disappointments are sorted worst-first; alternatives are then compared lexicographically on the sorted vectors. Here B wins because its largest disappointment (0.8) is below A ’s (0.9); had they tied, the second-largest entries would decide, and so on.

3.2. Admissible probes and disappointment

A probe $q : [0, 1]^m \rightarrow \mathbb{R}$ encodes a rational decision question. The family $\mathcal{Q}$ is explicitly declared.

Definition 1 (Admissible probe). *A probe q is admissible if it is non-decreasing in each r_i (monotonicity) and can be rescaled such that $q(\mathbf{0}) = 0$ and $q(\mathbf{1}) = 1$ (normalisation).*

Canonical probes include singletons $q_i(r) = r_i$, the grand mean $q_\mu(r) = \frac{1}{m} \sum_i r_i$, and the grand maximum $q_{\max}(r) = \max_i r_i$. Monotonicity ensures weak Pareto compatibility, while strict compatibility requires the family to *separate* criteria (e.g., by including all singleton probes). The multiplicity of probes in $\mathcal{Q}$ is a deliberate modelling choice, as repeated probes weight the lexicographic sorting.

For each probe $q \in \mathcal{Q}$, let $q^* = \min_{x \in A} q(r(x))$ and $q^w = \max_{x \in A} q(r(x))$ be its active extrema. If the active range $R_q = q^w - q^*$ is strictly positive, the *normalised disappointment* of x under q is

$$D_q(x) = \frac{q(r(x)) - q^*}{R_q} \in [0, 1]. \quad (2)$$

The regret certificate is the vector of these probe-wise disappointments.

3.3. The lexicimax order

Collect the disappointments $D(x) = (D_{q_1}(x), \dots, D_{q_K}(x))$ and sort them in descending order to obtain $D^\downarrow(x)$. The Leximax Universal-Regret order is defined as

$$x \preceq_{\text{LexUR}} y \iff D^\downarrow(x) \leq_{\text{lex}} D^\downarrow(y). \quad (3)$$

The method compares sorted regret vectors from worst coordinate downward until the first strict difference. The LexUR solution x^{LexUR} minimises this order, returning a certificate identifying the probes attaining the largest binding regrets.

In practice, a tolerance $\tau > 0$ absorbs estimation noise. We define W_τ as the set of τ -undominated alternatives. It acts as a numerical reporting device rather than a transitive equivalence class.

3.4. Interval bounds and stability

To address sensitivity to bound estimates, robustness requires evaluating regrets over a box of plausible ideal and nadir intervals $\mathcal{B}$. We define two fundamental stability sets: the *necessary interval winner* set $S_{\text{necessary}}^\tau = \bigcap_{b \in \mathcal{B}} W_\tau(b)$ (options winning under all bounds), and the *possible interval winner* set $S_{\text{possible}}^\tau = \bigcup_{b \in \mathcal{B}} W_\tau(b)$ (options winning under some bound). When bounds are highly uncertain, these stability classes should be reported instead of a single fragile point recommendation.

3.5. Foundational guarantees

Theorem 1 (Existence and completeness). *For any finite A and finite $\mathcal{Q}$, x^{LexUR} exists and exact $\preceq_{\text{LexUR}}$ is a complete preorder on A .*

The standard lexicographic operator guarantees a well-defined complete ranking on finite sets.

Theorem 2 (Weak and strict Pareto compatibility). *Let x strictly Pareto-dominate y and let every $q \in \mathcal{Q}$ be monotone non-decreasing in each r_i . Then $D_q(x) \leq D_q(y)$ for all q , so $x \preceq_{\text{LexUR}} y$. If $\mathcal{Q}$ separates criteria, then $x \prec_{\text{LexUR}} y$.*

Monotonicity alone prevents strict inversion but permits ties. Separation breaks the tie, strictly enforcing the Pareto principle.

Corollary 1 (Dominated exclusion). *If $x \in A$ is Pareto-dominated by some $y \in A$, and $\mathcal{Q}$ separates criteria, then x is not LexUR-optimal over A .*

Theorem 3 (Stability of the implemented winner). *Let $\hat{D}_q(x)$ estimate $D_q(x)$ with $\eta = \max_{x,q} |\hat{D}_q(x) - D_q(x)|$. Suppose the exact winner x^{LexUR} beats every rival y at the first deciding coordinate with margin Δ . If $2\eta < \min\{\Delta - \tau, \tau\}$, the tolerance-based winner is unchanged.*

This guarantees that small perturbations cannot flip the recommendation provided the true margin exceeds the estimation noise.

4. Computation

LexUR is primarily a candidate-set selection rule. We present the exact finite algorithm, followed by a brief summary of how it can be formulated directly on a continuous feasible set.

4.1. Exact algorithm for finite candidate sets

The finite-set procedure is summarised in Figure 4. With naive probe evaluation, producing a complete lexicographic ranking of all candidates costs $O(NKm + NK \log K + NK \log N)$, where N is the number of candidates, m is the number of criteria, and K is the number of probes.

Procedure: Exact finite-set LexUR ($\tau = 0$)

1. **Dominance filter:** Remove any Pareto-dominated $x \in A$.
2. **Normalisation:** Find active range z_i^*, z_i^{nad} and compute $r_i(x)$.
3. **Anchor bounds:** Find $q^* = \min q(r(x))$ and $q^w = \max q(r(x))$.
4. **Regret evaluation:** Compute $D_q(x) = (q(r(x)) - q^*) / (q^w - q^*)$.
5. **Certificate sorting:** Sort the labelled pairs $(D_q(x), q)$ descending to obtain $D^\downarrow(x)$.
6. **Exact selection:** Return every candidate attaining the lexicographic minimum.
7. **Numerical reporting:** Report the τ -stability class.

Figure 4: Finite-set LexUR procedure.

4.2. Continuous formulations

Proposition 1 (Front-free continuous computation). *Lexicographic minimisation of the sorted regret vector over a continuous set $\mathcal{X}$ is exactly equivalent to sequentially minimising the cumulative sums of the p largest disappointments, which admits an exact linearisation for structured problems.*

Remark 1 (Computational bottleneck). Under structured settings, the required anchors can be computed directly, so LexUR may be applied without explicit front generation. However, computing the per-probe anchor $q^w = \max_{x \in \mathcal{X}} q(r(x))$ requires maximising a convex function over a convex set. These $2K$ global anchor subproblems may be no cheaper than front generation in the general case. We provide the full continuous derivations in the appendix.

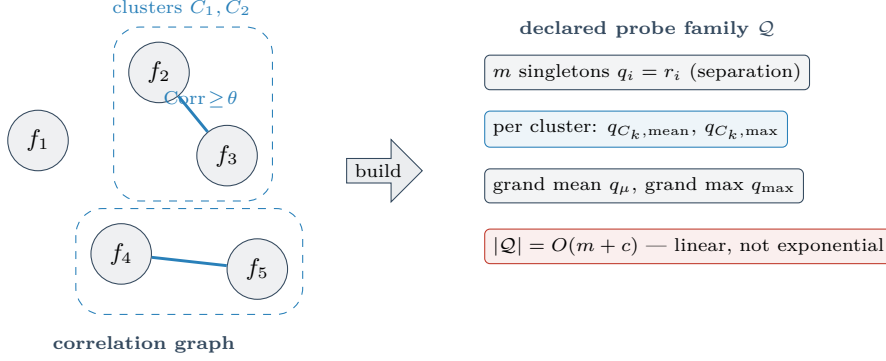


Figure 5: Adaptive probe construction. Positively correlated (redundant) criteria are grouped into clusters by thresholding the signed correlation at θ . Each nontrivial cluster contributes one mean and one max aggregate in place of the exponential subcoalition aggregates; singleton probes remain for separation. Together with the two grand probes, the family has $m + 2 + 2c$ probes, where c is the number of nontrivial clusters.

5. Adaptive, redundancy-aware probe generation

5.1. The redundancy problem and correlation clustering

Real criteria are often redundant, tracking the same underlying concern. Averaging selection rules implicitly over-weight such a group by counting it once per member. We propose an adaptive probe family that reduces this bias by clustering empirically redundant criteria.

We group redundant criteria by single-linkage connected components of the graph whose edges join pairs with Pearson correlation $\text{Corr}(f_i, f_j) \geq \theta$. Redundancy is positive correlation: two criteria that rise and fall together carry duplicate information, whereas negatively correlated criteria represent genuine trade-offs and must remain separate. This yields clusters $C_1, \dots, C_c$.

5.2. The adaptive probe family

LexUR always includes the m singletons (required for separation), the grand mean, and the grand maximum. It then adds one mean and one maximum probe *per cluster* of size > 1 :

$$\mathcal{Q} = \{q_i\}_{i=1}^m \cup \{q_\mu, q_{\max}\} \cup \bigcup_{|C_k|>1} \{q_{C_k, \text{mean}}, q_{C_k, \text{max}}\}. \quad (4)$$

Figure 5 illustrates the construction. A redundant group contributes one aggregate question rather than a separate aggregate per member. This reduces, but does not eliminate, redundancy bias because the individual singletons remain.

Theorem 4 (Pareto compatibility for adaptive probes). *If every probe is monotone, LexUR is weakly Pareto compatible. If the family contains the singleton probes, LexUR is strictly Pareto compatible.*

5.3. Perturbation bound

The reduction effect can be quantified as a regret-value perturbation lemma. This bounds the change in individual disappointments, not the change in the LexUR ranking.

Lemma 1 (Disappointment perturbation). *Suppose for a full-family probe q there is a reduced-family probe q' with $\sup_r |q(r) - q'(r)| \leq \eta$, and normalisation ranges satisfy $q^w - q^*, q'^w - q'^* \geq \beta > 0$. Then for every x :*

$$|D_q(x) - D_{q'}(x)| \leq \frac{4\eta}{\beta}.$$

The explicit proof is provided in the appendix. The bound is informative only when the probe's attainable range is bounded away from zero. Ultimately, correlation clustering is an empirical heuristic whose efficacy is demonstrated in §7.

Table 2: Probe-family size: full coalition family versus adaptive family.

m	full coalitions	adaptive probes	reduction
3	11	5	$2.2\times$
5	57	7	$8.1\times$
8	502	10	$50.2\times$
10	2,036	12	$169.7\times$
15	65,519	17	$3,854.1\times$

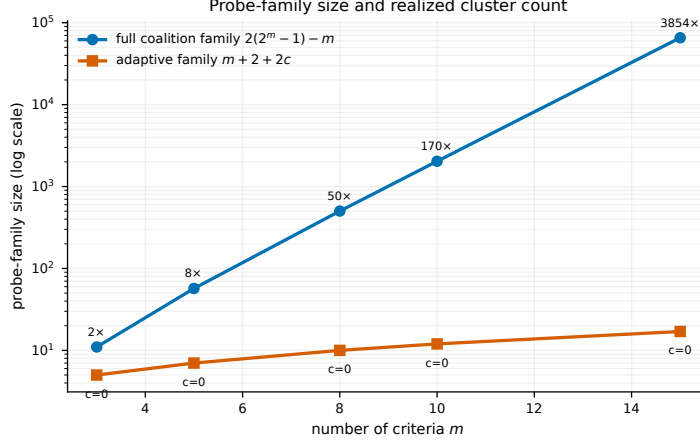


Figure 6: Probe-family size versus the number of criteria m .

5.4. Empirical reduction

Table 2 and Figure 6 report the realised reduction. The full coalition family has size $2(2^m - 1) - m$. The adaptive family is linear in m , achieving a nearly four-thousand-fold reduction by $m = 15$ while preserving decision quality.

6. Extensions

6.1. Stochastic regret certificates

When criteria are noisy, we model $f_i(x)$ as drawn from a joint distribution over scenarios. Each probe value is a random variable, and we apply a generic monotone risk functional ρ (e.g., sample average or Conditional Value-at-Risk) directly to the probe value $q(r(x, \omega))$. The normalisation anchors are taken consistently on this functional. We define the stochastic regret as

$$D_q^\rho(x) = \frac{\rho_q(x) - \rho_q^*}{\rho_q^w - \rho_q^*}. \quad (5)$$

For the empirical experiments in §7, we evaluate a specific *confidence-aware* variant using upper-confidence bounds $\tilde{f}_i(x) = \hat{\mu}_i(x) + z_\gamma \hat{\sigma}_i(x)/\sqrt{S}$ before applying the deterministic LexUR rule. While this heuristic provides practical risk control, the empirical standard deviation is not pointwise monotone, meaning exact scenario-wise Pareto compatibility is not strictly preserved by this specific statistical variant.

For theoretically strict risk control, a monotone functional like CVaR must be used instead. The stochastic framework preserves Pareto compatibility and recovers the true winner asymptotically under standard margin conditions, as formalised below (proofs in appendix).

Proposition 2 (Scenario-wise Pareto compatibility). *For finite A , finite $\mathcal{Q}$, and sample-average probe means, if x weakly dominates y in every scenario and strictly dominates it in at least one criterion for*

Table 3: Baseline selection rules used in §7.

Method	Score	Select	Parameters / tie-break
TOPSIS	relative closeness $d^-/(d^- + d^+)$	arg max	$w = \mathbf{1}/m$
CP	L_p distance to ideal	arg min	$w = \mathbf{1}/m, p = 2$
VIKOR	standard Q metric	arg min	$w = \mathbf{1}/m, v = 0.5$
Knee	distance below hyperplane through extremes	arg max	
HV ₁	single-point hypervolume from nadir	arg max	ref. 1
RW	frequency of arg min $w^\top r$ over random w	arg max	200 weights
ASF	augmented Chebyshev	arg min	$w = \mathbf{1}/m, \rho = 10^{-4}$
SMAA	first-rank acceptability under linear value	arg max	1000 weights
MMR	Savage regret	arg min	1000 weights
ChebMMR	Chebyshev minimax regret	arg min	1000 weights
LexUR	leximax of sorted normalised probe regrets	lex. min	$\theta = 0.6, \tau = 10^{-4}$

at least one scenario, then the sample-average LexUR rule does not prefer y to x . If $\mathcal{Q}$ separates criteria and the dominance is strict on a separated coordinate, x is strictly preferred.

Proposition 3 (Sample-average consistency). *If the population tolerance-LexUR winner satisfies the positive margin condition of Theorem 3, and population probe ranges are non-degenerate, then the sample-average LexUR winner converges in probability to the population winner as the sample size $S \rightarrow \infty$.*

6.2. Multi-stakeholder aggregation

While primarily a single decision-maker tool, LexUR can be extended to multi-stakeholder settings. If U indexes stakeholders, each assigning an importance weight to the criteria, one can define a composite family of stakeholder-specific probes. Lexicographic minimisation over this expanded family implements a Rawlsian maximin logic [8], protecting the most severely disappointed stakeholder on their most critical concerns. We leave the formal development and evaluation of this extension to future work.

7. Computational study

7.1. Design and protocol

All experiments are implemented in the accompanying modular Python reproducibility package. The evaluation protocol tests selection *quality* via a held-out analysis: each method selects one alternative using only its own logic, and the pick is scored against preference models that no method used during selection.

For a held-out utility u (higher is better) with in-sample best x_u^* and worst x_u^- , the held-out loss of a recommended alternative x is

$$L_u(x) = \frac{u(x_u^*) - u(x)}{u(x_u^*) - u(x_u^-)} \in [0, 1]. \quad (6)$$

We report the mean held-out loss and the tail held-out loss (CVaR_{10%}), emphasizing that robust methods exchange average performance to bound worst-case disappointment. Table 3 specifies the compared baselines.

7.2. Main Benchmark: The Broadened Protocol

To guard against confirmation bias, we ran a frozen audit protocol encompassing 11 methods, ten held-out preference families (linear, Chebyshev, augmented ASF, CES, Choquet, satisficing, sparse-linear, L_p , OWA, max-min), and eight front geometries. This generated 7,200 paired instances. The protocol fixes acceptance gates tied to specific claims, reported verbatim in Table 4.

Table 4: Frozen audit acceptance gates.

Claim	Gate	Result	Evidence
Pareto compatibility	Dominated selections = 0	PASS	0/150 injections
Scale invariance	Positive-affine invariance	PASS	100% identical
Stability under bounds	Nadir-error stability	CHECK	degr. 0.1533 (thr. 0.05)
Competitive w/ ASF	Non-inferior vs ASF (tail)	PASS	CI upper 0.0015 < 2% margin
Competitive w/ MMR	Non-inferior vs MMR (tail)	PASS	CI upper 0.0021 < 0.01 margin
Beats classical methods	Beats baselines (tail)	PASS	δ : TOPSIS .17, CP .07, SMAA .61
Redundancy correction	Redundancy < averaging	PASS	0.256 vs 0.359
Adaptive sufficiency	Adaptive $\approx$ full (tail gap $\leq$ 0.01)	PASS	gap 0.0053
Continuous applicability	Direct computation	PASS	19 vs 150. 4 vs 60 calls
Stochastic safety	Stochastic LexUR tail loss	PASS	risk 0.383 vs 0.383

Tail held-out loss ranks: frozen broadened protocol

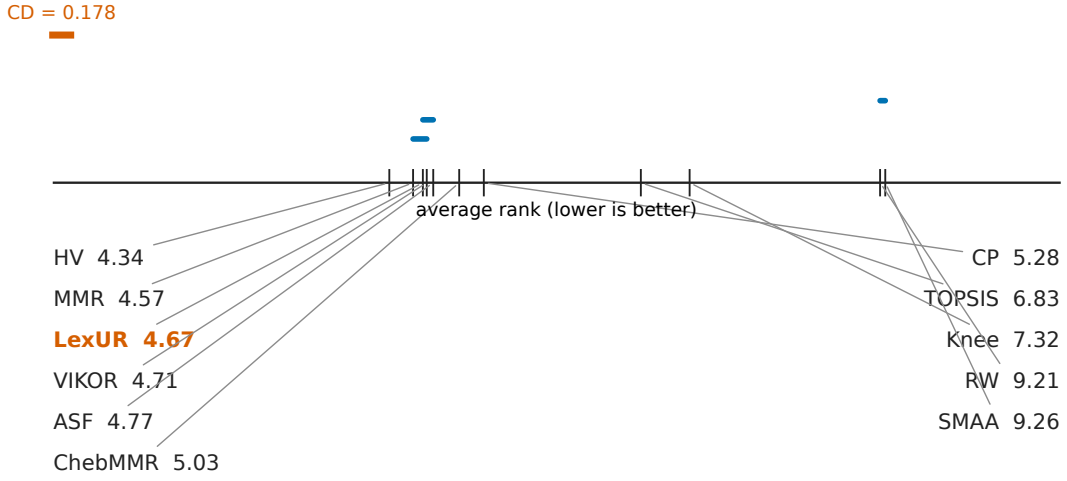


Figure 7: Average ranks on tail held-out loss over the broadened protocol suite.

LexUR achieves a tail-loss rank of 4.67. Within the Nemenyi critical difference, LexUR is statistically indistinguishable from the strongest robust scalarizations (MMR 4.57, VIKOR 4.71, ASF 4.77). Single-point HV achieves the best average rank (4.34), though it returns a bare point without semantic explanation. LexUR provides an auditable regret certificate while maintaining strict tail robustness.

The only CHECK outcome involves nadir-error stability, reinforcing that if the true nadir is unknown, practitioners must supply conservative interval bounds rather than relying on point estimates.

7.3. Mechanistic Analysis

To understand LexUR’s behaviour, we analyse a controlled 480-instance study. Table 5 and Figure 8 demonstrate the fundamental average-vs-worst-case tradeoff. LexUR groups with the strong robust methods (MMR, ASF, CP) on tail loss, clearly outperforming distance-based (TOPSIS) and popularity (RW, SMAA) rules on worst-case robustness.

Table 6 and Figure 10 detail performance across preference families. Averaging methods excel on linear utilities but fail on max-type profiles. ASF and MMR show the reverse. LexUR sits flexibly between them, providing strong baseline robustness without excessive tuning to a single preference structure.

When tested on redundant criteria (Table 7, Figure 11), LexUR suppresses redundancy bias effectively compared to aggregating baselines like TOPSIS and RW, validating the adaptive clustering strategy.

Table 5: Held-out loss over four geometries (480 instances).

Method	mean loss	tail loss
TOPSIS	0.220	0.539
CP	0.207	0.504
Knee	0.308	0.606
RW	0.462	0.900
ASF	0.234	0.472
SMAA	0.468	0.914
MMR	0.228	0.470
LexUR	0.245	0.481

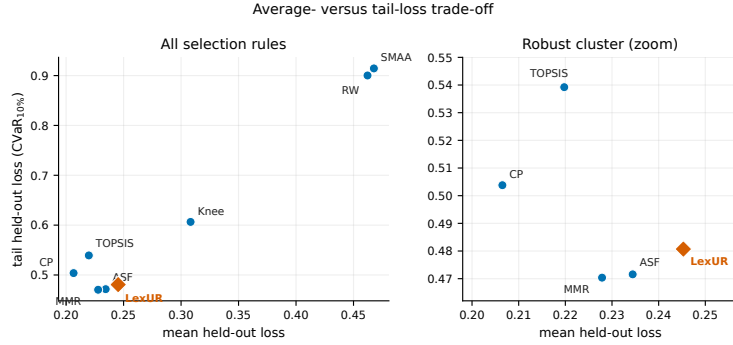


Figure 8: Average- versus worst-case held-out loss.

Ablation confirms the necessity of the complete probe family. Dropping singletons increases worst-case regret. Moreover, parameter sensitivity tests (Table 8, Figure 14) reveal LexUR is insensitive to the clustering threshold θ , but its selection identity can flip under nadir perturbations, again reinforcing the need for interval bounds.

7.4. Smart-grid dispatch case study

We applied LexUR to a six-objective stochastic economic dispatch, balancing cost, CO₂, unreliability, ramping wear, non-renewable generation, and peak-to-average gap. LexUR accepts a negligible increase in mean loss for a 23% reduction in worst-case loss compared to distance/averaging methods. Crucially, its certificate (Figure 15) highlights the exact binding vulnerabilities—peak-shaving and non-renewable fraction—offering an explicit explanation rather than an opaque point recommendation.

8. Discussion and Conclusion

LexUR provides a robust, certifiable decision rule for multicriteria analysis. While average-case methods like the weighted sum or SMAA excel when decision makers seek typical performance across assumed probability distributions, LexUR is designed for scenarios where the decision maker cannot state precise trade-offs and must defend a single robust choice against the worst-case disappointment over a declared family of rational interpretations.

It accomplishes this by turning the candidate set into a lexicographically optimal compromise, accompanied by an auditable certificate that explicitly names the most severely binding probes. The empirical evidence across extensive benchmark geometries, preference families, and noisy configurations demonstrates that LexUR matches the tail-loss robustness of the strongest scalarizations (like Savage minimax regret) while dramatically reducing bias from redundant criteria.

However, the framework has clear, declared limitations. Because LexUR normalises based on the empirical extremes of the candidate set, it violates Independence of Irrelevant Alternatives. Adding or

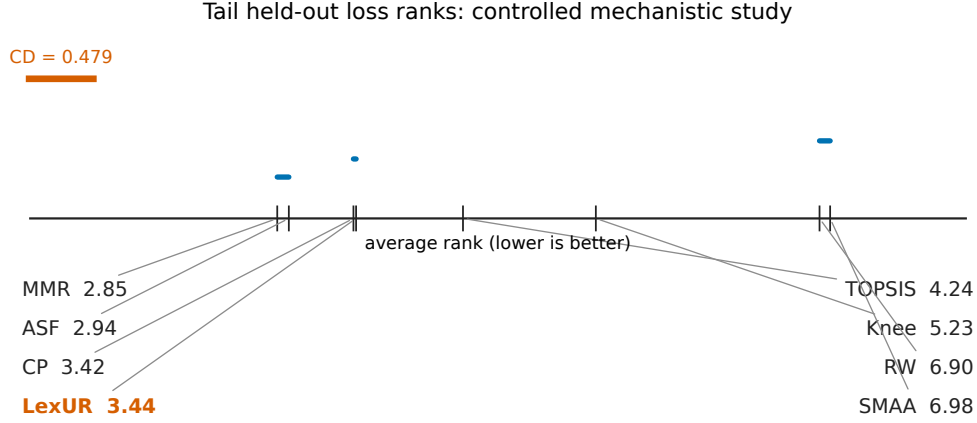


Figure 9: Average ranks on tail held-out loss with Nemenyi critical difference.

Table 6: Held-out loss by preference family.

Method	linear	Cheby.	aug. ASF	CES	Choquet	satisfice	overall
TOPSIS	0.391	0.511	0.511	0.563	0.341	0.511	0.471
CP	0.486	0.458	0.458	0.440	0.307	0.278	0.404
Knee	0.477	0.503	0.503	0.524	0.479	0.570	0.509
RW	0.403	0.519	0.519	0.569	0.379	0.552	0.490
ASF	0.674	0.434	0.434	0.329	0.288	0.388	0.425
SMAA	0.398	0.521	0.521	0.575	0.367	0.546	0.488
MMR	0.653	0.442	0.443	0.348	0.344	0.402	0.439
LexUR	0.598	0.454	0.454	0.374	0.368	0.468	0.453

removing unselected points can shift the active nadir, causing rank reversal in the final recommendation. Therefore, if the true normalisation bounds are uncertain, practitioners must employ interval estimates and report the resulting stability classes rather than relying on point-estimate selections. Additionally, LexUR trades average-case performance for its worst-case robustness.

Ultimately, LexUR replaces the traditional efficient set—a mere safety filter—with a certified recommendation. By relocating modelling choices into an explicit family of monotone questions, it gives decision makers an explainable, low-disappointment compromise that transparently reports its own fragility.

Reproducibility. All code, generated data, and scripts required to reproduce the protocol, the statistical tests, and the case study are released in an open-source repository. (URL withheld to preserve anonymity during review. It will be provided in the camera-ready version.)

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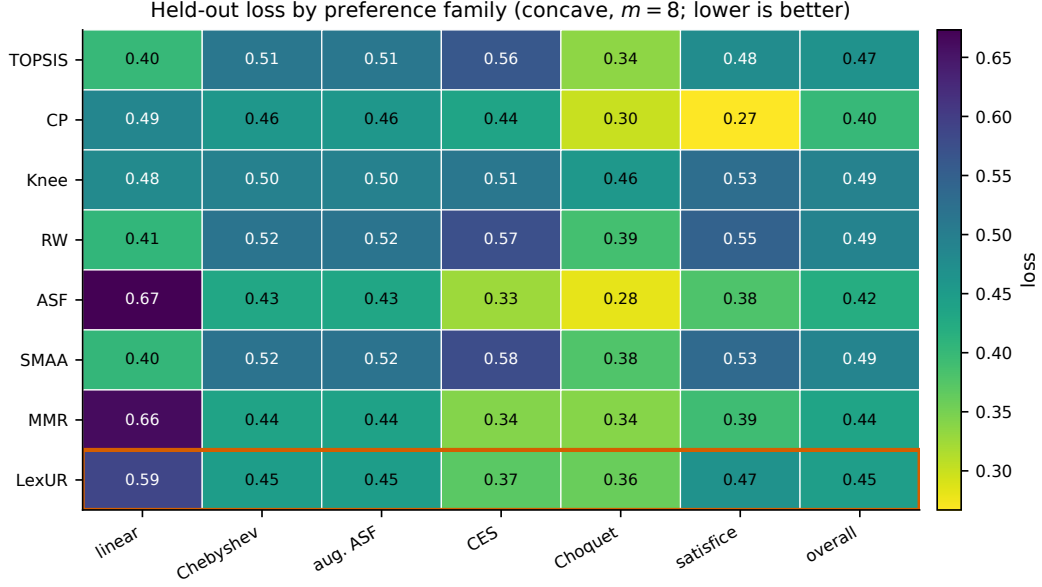


Figure 10: Held-out loss by preference family (concave, $m = 8$).

Table 7: Robustness to redundant criteria.

Method	grouped loss	tail loss
TOPSIS	0.332	0.455
CP	0.317	0.423
Knee	0.322	0.489
RW	0.409	0.598
ASF	0.266	0.406
SMAA	0.413	0.603
MMR	0.295	0.387
LexUR	0.268	0.391

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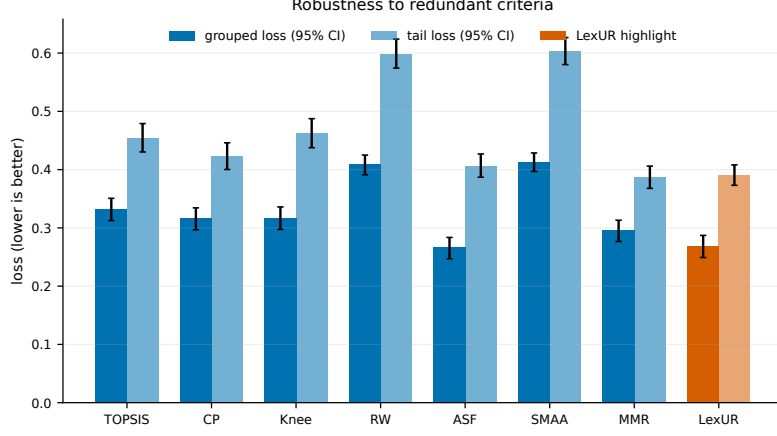


Figure 11: Robustness to redundant criteria.

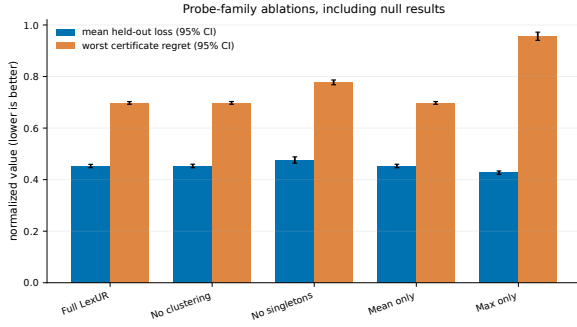


Figure 12: Ablation (concave, $m = 8$).

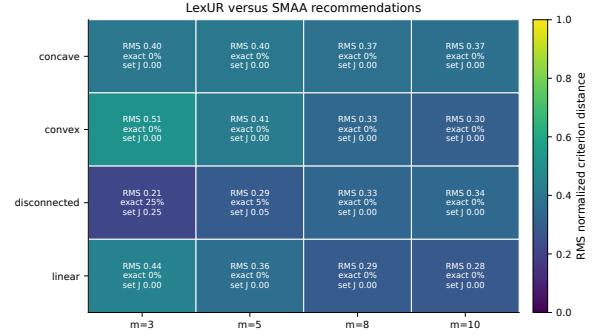


Figure 13: LexUR versus SMAA distinctness.

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Appendix A. Held-out preference families and broadened-protocol specification

This appendix gives the formal specification of the held-out evaluation families and the broadened audit protocol of §7.2, for reproducibility. None of these families is used to construct LexUR probes. they exist only to score recommendations out of class.

Notation and scoring.. Let $r(x) \in [0, 1]^m$ be the active-set normalised criterion vector of alternative x (minimisation, lower is better). Each family produces, for T sampled parameterisations, a utility $u(\cdot)$ (higher is better). a recommendation is scored by the held-out loss of Eq. (6), $L_u(x) = (u(x_u^*) - u(x))/(u(x_u^*) - u(x_u^-))$, with x_u^*, x_u^- the in-sample best and worst alternatives for u on A . Unless stated otherwise, importance weights are drawn from a Dirichlet distribution $w \sim \text{Dir}(\alpha \mathbf{1})$, whose shape parameter $\alpha \in \{0.2, 1, 5\}$ yields sparse/extreme, uniform-simplex, and balanced preferences respectively.

Table 8: Parameter diagnostics.

θ	linear	concave	convex	discon.	nadir err.	loss	flip rate
					0%	0.451	0%
0.3	0.179	0.459	0.036	0.351	5%	0.453	23%
0.5	0.179	0.452	0.036	0.355	10%	0.457	50%
0.8	0.176	0.452	0.037	0.360	20%	0.457	73%
					30%	0.458	90%
					50%	0.446	97%

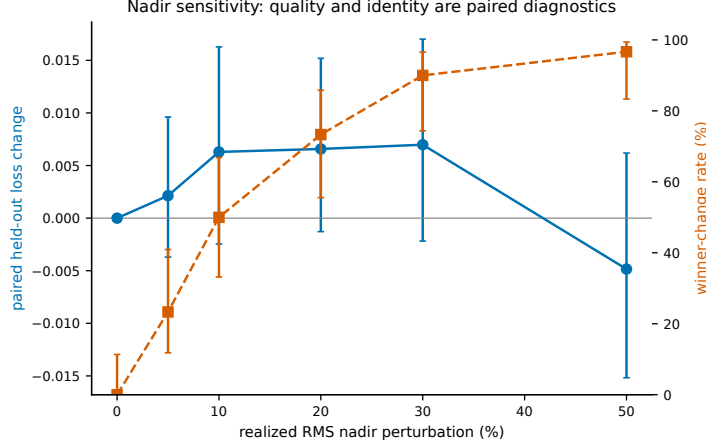


Figure 14: Nadir sensitivity.

The ten families.. Writing $u(x)$ for the utility (higher better), the families are:

- **linear:** $u(x) = -w^\top r(x)$, $w \sim \text{Dir}(\alpha \mathbf{1})$.
- **sparse-linear:** as linear but only the $k = \lceil m/3 \rceil$ largest weights are kept and renormalised to sum to one (a DM attending to a few criteria).
- **Chebyshev:** $u(x) = -\max_i w_i r_i(x)$ (weighted Chebyshev / pure ASF direction).
- **augmented ASF:** $u(x) = -(\max_i w_i r_i(x) + 0.05 w^\top r(x))$, i.e. the Chebyshev term plus a small $\rho = 0.05$ linear augmentation on the same weights.
- **L_p compromise:** $u(x) = -(\sum_i w_i r_i(x)^p)^{1/p}$ with exponent $p \sim \text{Unif}(1.5, 4.0)$ drawn per utility.
- **CES:** $u(x) = -(\sum_i w_i r_i(x)^\varrho)^{1/\varrho}$ with elasticity exponent $\varrho \sim \text{Unif}(2.0, 5.0)$ drawn per utility (constant-elasticity-of-substitution disutility). We note plainly that the L_p and CES families are *intentionally adjacent* rather than independent: both are norm-like disutilities of the same functional form, differing only in the exponent range. Likewise Chebyshev and augmented ASF are near-identical by construction (§7). We therefore do not claim ten *independent* preference geometries. the families are chosen to span a graded distance from the probe family (in-class to adversarial), not to be mutually orthogonal.
- **OWA:** $u(x) = -\sum_{k=1}^m \omega_k r_{[k]}(x)$, where $r_{[k]}$ is the k -th largest normalised criterion and the ordered weights $\omega = \text{sort}^\downarrow(\text{Dir}(\mathbf{1}))$ are a descending-sorted uniform-simplex draw (emphasis on worst coordinates).
- **Choquet-inspired (2-additive, max-interaction):** $u(x) = -(a^\top r(x) + \sum_{i < j} b_{ij} \max(r_i(x), r_j(x)))$ with non-negative masses $a_i \sim \text{Unif}(0, 1)$ on singletons and $b_{ij} \sim \text{Unif}(0, 0.5)$ on pairs. This is

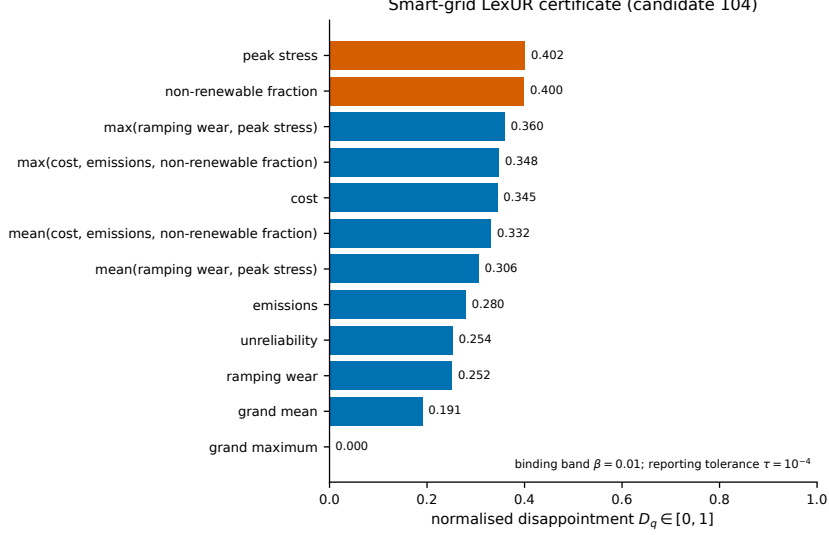


Figure 15: Complete LexUR regret certificate for smart-grid candidate.

a monotone non-additive pairwise-interaction utility in the *spirit* of a 2-additive Choquet integral. we deliberately avoid claiming it is exactly the standard 2-additive Choquet integral (whose Möbius/capacity form involves min terms for the pairwise interactions), and use it only as a genuinely non-additive held-out family.

- **satisficing:** $u(x) = -\sum_i \max(r_i(x) - \vartheta_i, 0)$ with per-criterion aspiration thresholds $\vartheta_i \sim \text{Unif}(0.2, 0.6)$, penalising only exceedances of the aspiration levels.
- **max-min (group regret):** three stakeholder weight vectors $w^{(1)}, w^{(2)}, w^{(3)} \sim \text{Dir}(\alpha \mathbf{1})$ define $u(x) = -\max_s (w^{(s)})^\top r(x)$, the worst-stakeholder linear disutility.

Families are grouped by distance from the probe family as *in-class* (linear, sparse-linear, Chebyshev, augmented ASF), *adjacent* (L_p , CES, OWA), *out-of-class* (Choquet, satisficing) and *adversarial* (max-min group regret). this scoping is fixed in the frozen config.

Protocol grid.. The broadened protocol of §7.2 sweeps candidate-set sizes $N \in \{100, 300, 1000\}$, criteria $m \in \{3, 5, 8, 10, 15, 20\}$, the eight geometries (linear, convex, concave, disconnected, degenerate, asymmetric, many-knee, irregular), and 50 replications, for $3 \times 6 \times 8 \times 50 = 7,200$ paired instances. A Dirichlet shape $\alpha \in \{0.2, 1, 5\}$ is assigned to each replication by deterministic cycling ($\alpha = \alpha_{\text{rep} \bmod 3}$), so the three shapes are nearly balanced (17, 17, 16 over the 50 replications) within each $(N, m, \text{geometry})$ cell. The paired non-inferiority confidence intervals (§7.2) use a cluster bootstrap with the instance keys as resampling units. The same candidate sets and held-out utility draws are shared across all methods within an instance to ensure paired comparisons. All parameter distributions above, the random seeds, and the frozen acceptance gates are fixed in the released configuration file and reproduced by a single command.

Front geometries.. Let s denote a point drawn on the unit simplex (symmetric Dirichlet) and d a vector with $d_i = \sqrt{U_i}$, $U_i \sim \text{Unif}(0, 1)$. every set is then reduced to its non-dominated subset. The eight geometries are:

- **linear:** $F = 0.5 s$ (hyperplane $\sum_i f_i = 0.5$).
- **convex:** $F = s^{\circ 2}$ (componentwise square. convex front $\sum_i \sqrt{f_i} = 1$).

- **concave:** $F = d/\|d\|_2$ (concave spherical front $\sum_i f_i^2 = 1$).
- **disconnected:** the concave sphere with the first coordinate offset by $0.6k$, $k \in \{0, 1, 2, 3\}$, opening gaps between patches.
- **degenerate:** the $s^{\circ 2}$ front with the last criterion tied to the first ($f_m \leftarrow f_1$), giving a lower-dimensional front.
- **asymmetric:** the concave sphere with criteria scaled by $\text{linspace}(1, 5, m)$ (heterogeneous criterion ranges).
- **many-knee:** the concave sphere warped by $F \mapsto F + 0.15 \sin(6F)$ to create multiple knees, then clipped to $F \geq 0$.
- **irregular:** the concave sphere with multiplicative noise $F \mapsto F \odot (1 + 0.1\mathcal{N}(0, 1))$, then clipped to $F \geq 10^{-3}$ so all criterion values stay non-negative.

All eight constructions yield non-negative criterion values (by clipping where noted), so the subsequent active-set normalisation maps them into $[0, 1]^m$.

Baseline specification.. Table 3 compactly specifies every comparator, all operating on the active-set normalised matrix $r \in [0, 1]^m$ (minimisation). w denotes equal weights $\mathbf{1}/m$ unless a method samples weights.

Appendix B. Smart-grid economic dispatch formulation

The case study in §7 evaluates a 24-hour multi-objective economic dispatch on a synthetic 14-generator instance constructed using the IEEE 14-bus demand/generator template. To prevent any misreading of the model’s fidelity, we describe the actual implementation precisely. it is deliberately a lightweight *priority-dispatch heuristic*, *not* a binary unit-commitment MILP and *not* a network-constrained AC/DC optimal power flow. It is an *aggregate* (single-bus) dispatch: the IEEE 14-bus data supplies the demand profile and the generator cost/emission/ramp template only, and no nodal power-flow, transmission-limit, line-loss, or binary commitment constraints are modelled (the dispatch balances the hourly demand $d_t(\omega)$ directly).

Candidate policy and dispatch rule.. A candidate policy is a non-negative *dispatch-priority* vector $\pi \in \mathbb{R}_{\geq 0}^{14}$, normalised to $\sum_g \pi_g = 1$. there are no binary commitment variables, and all generators are treated as available (renewable units have hour- and scenario-dependent available capacity). For each hour t and scenario ω , let $C_{gt}(\omega)$ be the available capacity ($C_{gt} = P_g^{\max}$ for thermal units, and the renewable cap scaled by the realised solar/wind availability for renewable units). The dispatch is computed by a two-pass priority rule:

$$p_{gt}(\omega) = \min(\pi_g d_t(\omega), C_{gt}(\omega)) \text{ (nominal share, capped),} \quad (\text{B.1})$$

after which any residual demand $d_t(\omega) - \sum_g p_{gt}(\omega)$ is filled greedily in proportion to the remaining headroom $C_{gt}(\omega) - p_{gt}(\omega)$, up to capacity. No ramp constraint is enforced. inter-hour ramping is *measured* as the wear objective below, not constrained. Renewable energy beyond what is dispatched is simply not used (there is no explicit curtailment variable). Any demand that still cannot be met is recorded as load shedding,

$$s_t^{\text{shed}}(\omega) = \max(d_t(\omega) - \sum_g p_{gt}(\omega), 0), \quad (\text{B.2})$$

which feeds the unreliability objective. Every sampled policy is therefore *feasible by construction*: capacity violations are impossible (dispatch is capped) and any supply shortfall is absorbed as load shedding rather than discarded or repaired.

The six objectives to be minimised are given below, each up to a fixed positive scaling constant that does not affect ranking or active-set normalisation. Let G_{th} and G_{ren} denote the thermal and renewable generator sets, c_g, e_g the fuel-cost and emission coefficients, and R_g the ramp scale.

1. **Cost:** $f_1(x) = \mathbb{E}_\omega \left[\sum_t \sum_g c_g p_{gt}(\omega) \right]$.
2. **Emissions:** $f_2(x) = \mathbb{E}_\omega \left[\sum_t \sum_g e_g p_{gt}(\omega) \right]$.
3. **Unreliability:** Loss-of-load probability, the fraction of scenarios in which any load is shed,

$$f_3(x) = \frac{1}{|\Omega|} \sum_{\omega \in \Omega} \mathbf{1} \left[\sum_t s_t^{\text{shed}}(\omega) > 0 \right].$$

4. **Ramping wear:** expected ramp magnitude normalised by the per-generator ramp scale R_g (a wear normaliser, *not* an enforced ramp limit),

$$f_4(x) = \mathbb{E}_\omega \left[\sum_g \sum_{t>1} \frac{|p_{g,t}(\omega) - p_{g,t-1}(\omega)|}{R_g} \right].$$

5. **Non-renewable fraction:** expectation of the per-scenario ratio of thermal to total dispatched

energy (an expectation of a ratio, not a ratio of expectations),

$$f_5(x) = \mathbb{E}_\omega \left[\frac{\sum_t \sum_{g \in G_{\text{th}}} p_{gt}(\omega)}{\sum_t \sum_g p_{gt}(\omega) + \varepsilon} \right].$$

6. **Peak stress:** a peak-shaving proxy equal to the peak demand scaled by the priority weight *not* pre-allocated to thermal generators,

$$f_6(x) = d^{\text{peak}} \max\left(1 - \sum_{g \in G_{\text{th}}} \pi_g, 0\right).$$

Each candidate policy is evaluated by Monte-Carlo sample-average approximation over 20 renewable scenarios drawn from the scenario model. The candidate set is generated *a priori*, independently of LexUR: we sample a population of 250 dispatch policies as Dirichlet-weighted generator allocations ($\text{Dir}(0.6 \cdot \mathbf{1})$ over the 14 generators, seed 2024), evaluate each under the scenarios, and retain the 163 non-dominated policies as the efficient candidate set. No evolutionary operators (selection, crossover, mutation) are used. a population metaheuristic such as NSGA-II could be substituted without changing LexUR’s role as a post-hoc selector. All parameter values (generator data, costs, emissions, ramp-wear coefficients R_g , and scenario bounds), random seeds, and the exact scenario count are fixed in the open-source repository. This formulation serves as an illustrative application of the robust certificate framework.

Appendix C. Multi-stakeholder (Rawlsian) exploratory variant

Let $\mathcal{U} = \{1, \dots, U\}$ index stakeholders, each with importance weights $w_u \in \Delta^m$ inducing a stakeholder-specific disappointment $D_q^u(x)$. An exploratory *Rawlsian LexUR* applies the leximax order to the worst-stakeholder disappointments, $\max_u D_q^u(x)$. It maintains natural fairness properties such as anonymity and worst-stakeholder protection. A full treatment of this extension is beyond the scope of this paper, but we include this preliminary formulation to demonstrate the flexibility of the regret-certificate framework.